

Ver A

PROBLEM 1. Given the vector field $\vec{F}(x, y, z) = x^2y \hat{i} + yz \cos x \hat{j} - x^2y^2e^z \hat{k}$.

- a. (2 points) Calculate divergence of $\vec{F}(x, y, z)$.
- b. (1 point) Describe whether the field is expanding, contracting or neither at these points.

i. $P = (\frac{1}{2}\pi, 1, 1)$

ii. $Q = (\pi, -1, 0)$

- c. (2 points) Calculate the curl of the vector field $\vec{F}(x, y, z)$.
- d. (5 point) Let $f(x, y, z)$ be a scalar function and $\vec{G}(x, y, z)$ a general vector field.
Show that $\vec{\nabla} \cdot (f\vec{G}) = f(\vec{\nabla} \cdot \vec{G}) + \vec{G} \cdot \vec{\nabla} f$

PROBLEM 2. (4 points) Find the work done due to the force field

$$\vec{F}(x, y, z) = (xyz)\hat{i} + (y^2 + zx)\hat{j} + (z + xy)\hat{k}$$

along the curve $C: \vec{r}(t) = t\hat{i} + 5t\hat{j} + 2t^2\hat{k}, -1 \leq t \leq 1$

PROBLEM 3. (5 point) Find the work done by $\vec{F}$ in moving a particle once counterclockwise around the given curve.

$$\vec{F} = 2xy^3\hat{i} + 4x^2y^2\hat{j}$$

C: The boundary of the "triangular" region in the first quadrant enclosed by the x -axis, the line $x = 1$ and the curve $y = x^3$.

PROBLEM 4.

- a. (2 points) Show that the function

$$f(x, y, z) = \frac{GmM}{\sqrt{x^2 + y^2 + z^2}}$$

is a potential function for the gravitational field

$$\vec{F}(x, y, z) = -GmM \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$$

Note: G, m and M are constants.

- b. (4 points) Let P_1 and P_2 be points at a distance s_1 and s_2 from the origin. Show that the work done by the gravitational field in part (a) in moving a particle from P_1 to P_2 is

$$GmM \left(\frac{1}{s_2} - \frac{1}{s_1} \right)$$

PROBLEM 5. (5 marks) Among all smooth, simple closed curves in the plane, oriented counterclockwise, find the one along which the work done by $F = \left(\frac{1}{4}x^2y + \frac{1}{12}y^3\right)\hat{i} + x\hat{j}$ is the maximum.

Hint: Find the region where $\text{curl} F \cdot \hat{k}$ is positive and explain what this has to do with the problem at hand.

End of Exam

"He who would learn to fly one day must first learn to stand and walk and run and climb and dance; one cannot fly into flying."

~Friedrich Nietzsche

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl} F \cdot \hat{k} \, dA$$



Problem 1

$$\vec{F}(x,y,z) = \overbrace{x^2y}^P \hat{i} + \overbrace{yz \cos x}^Q \hat{j} - \overbrace{x^2y^2e^z}^R \hat{k}$$

a) Divergence

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

(1 Point)

$$= 2xy + z \cos x - x^2y^2e^z$$

(1 Point)

b) i) $\vec{\nabla} \cdot \vec{F} \big|_{P=(\frac{\pi}{2}, 1, 1)} = \pi + \cos(\frac{\pi}{2}) - \frac{\pi^2}{4}e$

$$= \pi - \frac{\pi^2}{4}e$$

(1/2 Point)

~~in $\vec{F}$ at P~~

$$< 0$$

$\therefore \vec{F}$ is contracting at P

(0.5 Points)

ii) $\vec{\nabla} \cdot \vec{F} \big|_{Q=(\pi, -1, 0)} = -2\pi - \pi^2 < 0$

$\therefore \vec{F}$ is contracting at Q

(0.5 Points)

c) Curl

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & yz \cos x & -x^2y^2e^z \end{vmatrix}$$

(1 Point)

$$= [-2x^2ye^z \hat{i} - \{y \cos x\} \hat{j} - [-2xy^2e^z] \hat{j} + [-yz \sin x - x^2] \hat{k}]$$

(1 Point)

$$= -(2x^2ye^z + y \cos x) \hat{i} + 2xy^2e^z \hat{j} - (yz \sin x - x^2) \hat{k}$$

If someone has not written curl as a vector then (1/2 point)

d) $f(x, y, z)$ scalar function

$\vec{G}(x, y, z)$ a vector field.

Prove that $\vec{\nabla} \cdot (f\vec{G}) = f(\vec{\nabla} \cdot \vec{G}) + \vec{G} \cdot \vec{\nabla} f$

Let $\vec{G}(x, y, z) = P(x, y, z)\hat{i} + Q(x, y, z)\hat{j} + R(x, y, z)\hat{k}$ — (0.5)

Then $f\vec{G} = f(x, y, z)P(x, y, z)\hat{i} + f(x, y, z)Q(x, y, z)\hat{j} + f(x, y, z)R(x, y, z)\hat{k}$

LHS: $\vec{\nabla} \cdot (f\vec{G}) = \frac{\partial}{\partial x}(fP) + \frac{\partial}{\partial y}(fQ) + \frac{\partial}{\partial z}(fR)$ — (0.5)

By Product Rule of derivatives

$$\vec{\nabla} \cdot (f\vec{G}) = \frac{\partial f}{\partial x}P + f\frac{\partial P}{\partial x} + \frac{\partial f}{\partial y}Q + f\frac{\partial Q}{\partial y} + \frac{\partial f}{\partial z}R + f\frac{\partial R}{\partial z}$$
 — (2)

Rearranging, we get

$$\textcircled{1} - = \cancel{\frac{\partial f}{\partial x}P} + f\left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) + \left[\frac{\partial f}{\partial x}P + \frac{\partial f}{\partial y}Q + \frac{\partial f}{\partial z}R\right]$$

$$= f(\vec{\nabla} \cdot \vec{G}) + \vec{\nabla} f \cdot \vec{G}$$

Since dot product is commutative

$$= f(\vec{\nabla} \cdot \vec{G}) + \vec{G} \cdot \vec{\nabla} f \quad \text{--- (1)}$$

= R.H.S



PROBLEM 2

$$\vec{F}(x,y,z) = (xyz)\hat{i} + (y^2+zx)\hat{j} + (z+xy)\hat{k}$$

$$C: \vec{r}(t) = t\hat{i} + 5t\hat{j} + 2t^2\hat{k}, \quad -1 \leq t \leq 1$$

Compute $\int_C \vec{F} \cdot d\vec{r}$

$$\vec{r}'(t) = \hat{i} + 5\hat{j} + 4t\hat{k} \quad \text{--- (0.5)}$$

$$\vec{F}(\vec{r}(t)) = 10t^4\hat{i} + (25t^2 + 2t^3)\hat{j} + (2t^2 + 5t^2)\hat{k}$$

$$\vec{F}(t) = 10t^4\hat{i} + (25t^2 + 2t^3)\hat{j} + (7t^2)\hat{k} \quad \text{--- (1)}$$

$$\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) = 10t^4 + 125t^2 + 10t^3 + 28t^3$$

$$= 125t^2 + 38t^3 + 10t^4 \quad \text{--- (1)}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_{-1}^1 \vec{F} \cdot \vec{r}'(t) dt$$

$$= \int_{-1}^1 (125t^2 + 38t^3 + 10t^4) dt \quad \text{--- (0.5)}$$

$$= \left. \frac{125}{3}t^3 + \frac{19}{2}t^4 + 2t^5 \right|_{-1}^1 \quad \text{--- (0.5)}$$

$$= \frac{125}{3} \times 2 + 2 \times 2$$

$$= \frac{262}{3} \quad \text{--- (0.5)}$$

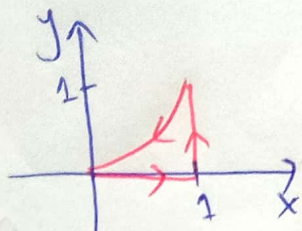
For correct limits
and symbols

Problem 3:

$$\vec{F} = 2xy^3 \hat{i} + 4x^2y^2 \hat{j}$$

$C :=$ ~~boundary~~ boundary of region b/w $x=1, y=0$
 & $y=x^3$. oriented counter clock wise.

Solution: Sketch



Since C is piecewise smooth, closed, simple & positively oriented curve and $\vec{F}$ has continuous first partial derivatives everywhere, we can apply the Green's Theorem. -0.5

Green's Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl } \vec{F} \cdot \hat{k} \, dA \quad \text{where } R \text{ is region enclosed by } C. \quad \text{--0.5}$$

1 mark
for correct
limits & symbols

$$= \int_{x=0}^1 \int_{y=0}^{y=x^3} (8xy^2 - 6xy^2) \, dy \, dx \quad \text{--1}$$

$$= \int_{x=0}^1 \int_{y=0}^{y=x^3} 2xy^2 \, dy \, dx$$

$$= \int_0^1 \left. \frac{2}{3} xy^3 \right|_{y=0}^{y=x^3} dx \quad \text{--1}$$

$$= \int_0^1 \frac{2}{3} x^{10} \, dx$$

$$= \frac{2}{33} \quad \text{--1}$$

If some one has found work done by line integrals using correct parametrisation for all curves then they get full score.

Problem 4

$$f = \frac{G_m M}{\sqrt{x^2 + y^2 + z^2}}$$

$$\vec{F} = -G_m M \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$$

a) Show that f is a potential function of $\vec{F}$

f is a potential function of $\vec{F}$ iff $\vec{\nabla} f = \vec{F}$ (0.5)

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$\text{Computing } \frac{\partial f}{\partial x}: \quad \frac{\partial f}{\partial x} = G_m M \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} \\ = G_m M \left(-\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x) \right)$$

$$= -G_m M \frac{x}{(x^2 + y^2 + z^2)^{3/2}} \quad \text{--- (1)}$$

Since f is symmetric in x, y & z

$$\frac{\partial f}{\partial y} = -G_m M \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\text{& } \frac{\partial f}{\partial z} = -G_m M \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\text{Thus } \vec{\nabla} f = -G_m M \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} = \vec{F}$$

(0.5)

□

Problem 5: find C such that $\oint_C \vec{F} \cdot d\vec{r}$ is maximum

where $\vec{F} = \left(\frac{1}{4}x^2y + \frac{1}{12}y^3 \right) \hat{i} + x\hat{j}$

and C smooth, simple, closed, oriented counterclockwise.

Want C such that $\max_C \left(\oint_C \vec{F} \cdot d\vec{r} \right)$

① Since all the conditions of h-T are satisfied

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl} \vec{F} \cdot \hat{k} \, dA \quad \text{where } R \text{ is the region enclosed by } C.$$

Thus work done will be maximum, when the integral on the right is maximum - 0.5

① Integral ~~increases~~ increases if the integrand is positive and decreases when the integrand is negative.

Thus to maximize the integral we need a maximum R such that $\text{curl} \vec{F} \cdot \hat{k}$ is positive over this R .

$$\text{curl} \vec{F} \cdot \hat{k} = 1 - \left(\frac{1}{4}x^2 + \frac{1}{4}y^2 \right) \quad - 1$$

$$1 - \left(\frac{1}{4}x^2 + \frac{1}{4}y^2 \right) > 0 \quad \text{when}$$

$$4 > x^2 + y^2$$

Thus $\max_R$ is a disk centered at origin with radius 2. - 1

and C is the boundary circle $x^2 + y^2 = 2$ - 0.5

Ver B

PROBLEM 1. Given the vector field $\vec{F}(x, y, z) = xy^2 \hat{i} + yz \cos x \hat{j} - x^2 yz^2 \hat{k}$.

- a. (2 points) Calculate divergence of $\vec{F}(x, y, z)$.
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i. $P = (\frac{1}{2}\pi, -1, 1)$

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PROBLEM 2. (4 points) Find the work done due to the force field

$$\vec{F}(x, y, z) = (x + yz)\hat{i} + (y + zx)\hat{j} + (z + xy)\hat{k}$$

along the curve $C: \vec{r}(t) = t\hat{i} + 5t\hat{j} + 2t^2\hat{k}, -1 \leq t \leq 1$

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$$\vec{F} = 3xy^2\hat{i} + 2x^3y^2\hat{j}$$

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Note: G, m and M are constants.

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$$GmM \left(\frac{1}{s_2} - \frac{1}{s_1} \right)$$

Ver B

PROBLEM 5. (5 marks) Among all smooth, simple closed curves in the plane, oriented counterclockwise, find the one along which the work done by $F = \left(\frac{1}{4}x^2y + \frac{1}{12}y^3\right)\hat{i} + x\hat{j}$ is the maximum.

Hint: Find the region where $\text{curl} F \cdot \hat{k}$ is positive and explain what this has to do with the problem at hand.

End of Exam

"He who would learn to fly one day must first learn to stand and walk and run and climb and dance; one cannot fly into flying."

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VER B

Problem 1

$$\vec{F} = \frac{P}{xy^2} \hat{i} + \frac{Q}{yz \cos x} \hat{j} - \frac{R}{x^2 y z^2} \hat{k}$$

a) Divergence:

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \quad - (1)$$

$$= y^2 + z \cos x - 2x^2 y z \quad - (1)$$

b) i) $\vec{\nabla} \cdot \vec{F} \Big|_{P = (\frac{1}{2}\pi, -1, 1)} = 1 + 0 - 2\left(\frac{1}{2}\pi\right)^2 (-1)(1)$

$$= 1 + \frac{\pi^2}{2} > 0$$

Since $\vec{\nabla} \cdot \vec{F} \Big|_P > 0$, $\vec{F}$ is expanding at P (0.5 points)

ii) $\vec{\nabla} \cdot \vec{F} \Big|_{Q = (\pi, 1, \pi^2)} = 1 + \pi^2 (-1) - 2(\pi^2)(1)\pi^2$

< 0
 $\therefore \vec{F}$ is contracting at Q (0.5 Points)

c) $\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & yz \cos x & -x^2 y z^2 \end{vmatrix}$ (1 point)

$$= (-x^2 z^2 - y \cos x) \hat{i} - (-2xyz^2 - 2xy) \hat{j} + (-yz \sin x - 2xy) \hat{k}$$

$$= (-x^2 z^2 - y \cos x) \hat{i} + 2xyz^2 \hat{j} + (-yz \sin x - 2xy) \hat{k} \quad (1 \text{ point})$$

If someone has not written curl as a vector then (1/2) points

d) $f(x, y, z)$ scalar function

$\vec{G}(x, y, z)$ a vector field.

Prove that $\vec{\nabla} \cdot (f\vec{G}) = f(\vec{\nabla} \cdot \vec{G}) + \vec{G} \cdot \vec{\nabla} f$

Let $\vec{G}(x, y, z) = P(x, y, z)\hat{i} + Q(x, y, z)\hat{j} + R(x, y, z)\hat{k}$ — (0.5)

Then $f\vec{G} = f(x, y, z)P(x, y, z)\hat{i} + f(x, y, z)Q(x, y, z)\hat{j} + f(x, y, z)R(x, y, z)\hat{k}$ — (0.5)

LHS: $\vec{\nabla} \cdot (f\vec{G}) = \frac{\partial}{\partial x}(fP) + \frac{\partial}{\partial y}(fQ) + \frac{\partial}{\partial z}(fR)$

By Product Rule of derivatives

$$\vec{\nabla} \cdot (f\vec{G}) = \frac{\partial f}{\partial x}P + f\frac{\partial P}{\partial x} + \frac{\partial f}{\partial y}Q + f\frac{\partial Q}{\partial y} + \frac{\partial f}{\partial z}R + f\frac{\partial R}{\partial z}$$
 — (2)

Rearranging, we get

$$\begin{aligned} \textcircled{1} - &= \cancel{\frac{\partial f}{\partial x}} f \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) + \left[\frac{\partial f}{\partial x}P + \frac{\partial f}{\partial y}Q + \frac{\partial f}{\partial z}R \right] \\ &= f(\vec{\nabla} \cdot \vec{G}) + \vec{\nabla} f \cdot \vec{G} \end{aligned}$$

Since dot product is commutative

$$= f(\vec{\nabla} \cdot \vec{G}) + \vec{G} \cdot \vec{\nabla} f \quad \text{--- (1)}$$

= R.H.S



Problem 2 $\vec{F}(x,y,z) = (x+yz)\hat{i} + (y+zx)\hat{j} + (z+xy)\hat{k}$
 $C: \vec{r}(t) = t\hat{i} + 5t\hat{j} + 2t^2\hat{k}, -1 \leq t \leq 1$

Compute $\int_C \vec{F} \cdot d\vec{r}$

$$\vec{r}'(t) = \langle 1, 5, 4t \rangle \quad - (0.5)$$

$$\vec{F}(\vec{r}(t)) = \langle t+10t^3, 5t+2t^3, 7t^2 \rangle \quad - (1)$$

$$\begin{aligned} \vec{F}(t) \cdot \vec{r}'(t) &= t+10t^3+25t+10t^3+28t^3 \\ &= 26t+20t^3+28t^3 \\ &= 26t+48t^3 \quad - (1) \end{aligned}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_{-1}^1 (26t+48t^3) dt \quad - (0.5)$$

for correct limits & symbols.

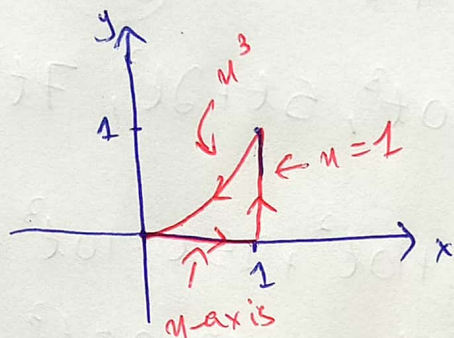
$$= (13t^2+12t^4) \Big|_{-1}^1 \quad - (0.5)$$

$$= 0 \quad - (0.5)$$

Problem 3: $\vec{F} = 3xy^2 \hat{i} + 2x^3y^2 \hat{j}$

C : Boundary of the "triangular" region bounded by x -axis, $x=1$ & $y=x^3$, oriented counter clockwise.

Solution: Sketch



Since the curve is positively oriented, closed, simple & piecewise smooth & $\vec{F}$ has continuous first partial derivatives everywhere, we can apply the Green's Theorem.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl} \vec{F} \cdot \hat{k} \, dA$$

where R is the region enclosed by C .

①.5 for reasoning why Green's Thm can be used.

①.5 for using U.T.

① for correct order of integration & limits.

$$= \int_0^1 \int_0^{x^3} (6x^2y^2 - 6xy) \, dy \, dx$$

$$= \int_0^1 2x^2y^3 - 3xy^2 \Big|_{y=0}^{y=x^3} \, dx$$

$$= \int_0^1 2x^8 - 3x^7 \, dx$$

$$= \left. \frac{1}{6}x^9 - \frac{3}{8}x^8 \right|_0^1 = \frac{1}{6} - \frac{3}{8} = -\frac{5}{24}$$

If someone has found work done directly by line integrals, they get the correct solution.

Problem 4

$$f = \frac{G_m M}{\sqrt{x^2 + y^2 + z^2}}$$

$$\vec{F} = -G_m M \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$$

a) Show that f is a potential function of $\vec{F}$

f is a potential function of $\vec{F}$ iff $\vec{\nabla} f = \vec{F}$ (0.5)

$$\vec{\nabla} f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$\text{Computing } \frac{\partial f}{\partial x}: \quad \frac{\partial f}{\partial x} = G_m M \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} \\ = G_m M \left(-\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x) \right)$$

$$= -G_m M \frac{x}{(x^2 + y^2 + z^2)^{3/2}} \quad (1)$$

Since f is symmetric in x, y & z

$$\frac{\partial f}{\partial y} = -G_m M \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\text{& } \frac{\partial f}{\partial z} = -G_m M \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\text{Thus } \vec{\nabla} f = -G_m M \frac{(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}} = \vec{F}$$

(0.5)

□

b) P_1 & P_2 two points at distance s_1 & s_2 away from origin. Show that work done by $\vec{F}$ from P_1 to P_2 is $GmM \left(\frac{1}{s_2} - \frac{1}{s_1} \right)$.

Let $P_1 = (x_1, y_1, z_1)$ then $s_1 = \sqrt{x_1^2 + y_1^2 + z_1^2}$ } ①

Let $P_2 = (x_2, y_2, z_2)$ then $s_2 = \sqrt{x_2^2 + y_2^2 + z_2^2}$

Since $\vec{F}$ is a gradient field from (part a)

$\vec{F}$ is conservative and by fundamental theorem of Vector Calculus } ②

$$\int_{P_1}^{P_2} \vec{F} \cdot d\vec{r} = f(P_2) - f(P_1)$$

$$= \frac{GmM}{\sqrt{x_2^2 + y_2^2 + z_2^2}} - \frac{GmM}{\sqrt{x_1^2 + y_1^2 + z_1^2}}$$

$$= \frac{GmM}{s_2} - \frac{GmM}{s_1} \quad \text{--- ①}$$

$$= GmM \left(\frac{1}{s_2} - \frac{1}{s_1} \right)$$

□

Problem 5: find C such that $\oint_C \vec{F} \cdot d\vec{r}$ is maximum

where $\vec{F} = \left(\frac{1}{4} x^2 y + \frac{1}{12} y^3 \right) \hat{i} + x \hat{j}$

and C smooth, simple, closed, oriented counterclockwise.

Want C such that $\max_C \left(\oint_C \vec{F} \cdot d\vec{r} \right)$

① Since all the conditions of h-T are satisfied

① $\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl} \vec{F} \cdot \hat{k} \, dA$ where R is the region enclosed by C . ~~⊗~~

Thus work done will be maximum, when the integral on the right is maximum - 0.5

① Integral ~~in maximum~~ increases if the integrand is positive and decreases when the integrand is negative.

① Thus to maximize the integral we need a maximum R such that $\text{curl} \vec{F} \cdot \hat{k}$ is positive over this R .

$$\text{curl} \vec{F} \cdot \hat{k} = 1 - \left(\frac{1}{4} x^2 + \frac{1}{4} y^2 \right) \quad \text{--- ①}$$

$$1 - \left(\frac{1}{4} x^2 + \frac{1}{4} y^2 \right) > 0 \quad \text{when}$$

$$4 > x^2 + y^2$$

Thus $\max_R$ is a disk centered at origin with radius 2. --- ①

and C is the boundary circle $x^2 + y^2 = 2$ --- 0.5